

# **ABSTRACT**

## **Generative AI and Large Language Models in Supply Chain Management: Opportunities, Risks, and Future Directions**

The emergence of Generative Artificial Intelligence (Generative AI) and Large Language Models (LLMs) marks a significant technological advancement that is reshaping the landscape of supply chain management. Traditional Artificial Intelligence applications in supply chains have primarily focused on prediction, optimization, and automation using structured data. In contrast, Generative AI extends these capabilities by enabling machines to understand, generate, summarize, and interact with human language, thereby unlocking the value hidden within vast amounts of unstructured data such as contracts, emails, invoices, supplier reports, and operational documents.

This seminar examines the role of Generative AI and LLMs in enhancing various supply chain functions, including procurement, demand forecasting, inventory management, logistics, warehousing, supplier relationship management, and risk mitigation. The ability of these technologies to provide intelligent recommendations, generate reports, assist in decision-making, and support real-time communication has the potential to improve operational efficiency, visibility, and responsiveness across the entire supply chain network.

The seminar further explores how Generative AI can contribute to building more resilient supply chains by supporting scenario analysis, disruption monitoring, and contingency planning in response to global uncertainties such as pandemics, geopolitical conflicts, and transportation disruptions. At the same time, important challenges associated with the adoption of these technologies are discussed, including hallucinations, data privacy concerns, bias, cybersecurity risks, and issues related to transparency and explainability.

Finally, the seminar highlights the future direction of AI-enabled supply chains, emphasizing the growing importance of human-AI collaboration and the development of intelligent decision-support systems and autonomous supply chain operations. The study concludes that while Generative AI and LLMs are unlikely to replace supply chain professionals, they will fundamentally transform the way supply chain decisions are made, creating new opportunities for innovation, agility, and competitive advantage in modern organizations.